PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 50-
ANNEX 4/Sheet 3
REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT AREAS OF SUBSTANTIAL CORROSION RENEWAL SURVEY OF DOUBLE HULL OIL TANKERS WITHIN THE CARGO AREA LENGTH
STRUCTURE IN WING BALLAST TANKS
Structural member Extent of measurement Pattern of measurement
Side shell and longitudinal bulkhead plating:
- upper strake and strakes in way of horizontal girders Single measurement
Plating between each pair of longitudinals in a minimum of three bays (along the tank)
- upper strake and strakes Plating between every third Single measurement
- all other strakes pair of longitudinals in same three bays
Side shell and longitudinal bulkhead longitudinals on: - upper strake - all other strakes Each longitudinal in same three bays 3 measurements across web flange
Longitudina ls - brackets Minimum of three at top, middle and bottom of tank in bracket
Vertical web and transverse bulkheads (excluding deckhead area): - strakes in way of horizontal girders - other strakes Minimum of two webs and 5-point pattern over approx. two square metre area
Minimum of two webs and two measurements between both transverse bulkheads each pair of vertical stiffeners
Horizontal girders Plating on each girder in a minimum of three bays Two measurements between each pair of longitudinal girder stiffeners
Panel stiffening Where applicable Single measurements

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SOURCE: MSC.197(80).pdf
STRUCTURE IN WING BALLAST TANKS
Structural member Extent of measurement Pattern of measurement
Side shell and longitudinal bulkhead plating:
- upper strake and strakes in way of horizontal girders Single measurement
Plating between each pair of longitudinals in a minimum of three bays (along the tank)
- upper strake and strakes Plating between every third Single measurement
- all other strakes pair of longitudinals in same three bays
Side shell and longitudinal bulkhead longitudinals on: - upper strake - all other strakes Each longitudinal in same three bays 3 measurements across web flange
Longitudina ls - brackets Minimum of three at top, middle and bottom of tank in bracket
Vertical web and transverse bulkheads (excluding deckhead area): - strakes in way of horizontal girders - other strakes Minimum of two webs and 5-point pattern over approx. two square metre area
Minimum of two webs and two measurements between both transverse bulkheads each pair of vertical stiffeners
Horizontal girders Plating on each girder in a minimum of three bays Two measurements between each pair of longitudinal girder stiffeners
Panel stiffening Where applicable Single measurements

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SOURCE: MSC.197(80).pdf
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS

AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 51 -
ANNEX 4/Sheet 4
REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT AREAS OF SUBSTANTIAL CORROSION RENEWAL SURVEY OF DOUBLE HULL OIL TANKERS WITHIN THE CARGO AREA LENGTH
LONGITUDINAL BULKHEADS IN CARGO TANKS
Structural member Extent of measurement Pattern of measurement
Deckhead and bottom strakes, Plating between each pair of Single measurement
and strakes in way of the longitudinals in a minimum of
horizontal stringers of three bays
transverse bulkhead s
All other strakes Plating between every third Single measurement
pair of longitudinals in same
three bays

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SOURCE: MSC.197(80).pdf
and strakes in way of the longitudinals in a minimum of
horizontal stringers of three bays
transverse bulkhead s
All other strakes Plating between every third Single measurement
pair of longitudinals in same
three bays
Longitudinals on deckhead Each longitudinal in same Three measurements across
and bottom strakes three bays web and one measurement on
flange
All other longitudinals Every third longitudinal in Three measurements across
same three bays web and one measurement on
flange
Longitudinals - brackets Minimum of three at top, 5-point pattern over area of
middle and bottom of tank in bracket
same three bays
Web frames and cross ties Three webs with minimum of 5-point pattern over
three locations on each web, approximately two square
including in way of cross tie metre area of webs, plus
connections single measurements on
flanges of web frame and
cross ties
Lower end brackets Minimum of three brackets 5-point pattern over
(opposite side of web approximately two square
frame) metre area of brackets, plus
single measurements on
bracket flanges
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED

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SOURCE: MSC.197(80).pdf
(opposite side of web approximately two square
frame) metre area of brackets, plus
single measurements on

bracket flanges
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 52-
ANNEX 4/Sheet 5
REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT
AREAS OF SUBSTANTIAL CORROSION RENEWAL SURVEY OF DOUBLE
HULL OIL TANKERS WITHIN THE CARGO AREA LENGTH
TRANSVERSE WATERTIGHT AND SWASH BULKHEADS IN
CARGO TANKS
Structural member Extent of measurement Pattern of measurement
Upper and lower stool, where Transverse band within 5-point pattern between
fitted 25 mm of welded connection stiffeners over one metre
to inner bottom/deck plating length
Transverse band within
25 mm of welded connection
to shelf plate
Deckhead and bottom strakes, Plating between pair of 5-point pattern between
and strak es in way of stiffeners at three locations: stiffeners over one metre
horizontal stringers approximately 1/4, 1/2 and 3/4 length
width of tank
All other strakes Plating between pair of Single measurement

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SOURCE: MSC.197(80).pdf
and strak es in way of stiffeners at three locations: stiffeners over one metre
horizontal stringers approximately 1/4, 1/2 and 3/4 length
width of tank
All other strakes Plating between pair of Single measurement
stiffeners at middle location
Strakes in corrugated Plating for each change of 5-point pattern over about one
bulkheads scantling at centre of panel square metre of plating
and at flange of fabricated
connection
Stiffeners Minimum of three typical For web, 5-point pattern over
stiffeners span between bracket
connections (two
measurements across web at
each bracket connection and
one at centre of span).
For flange, single
measurements at each bracket
toe and at centre of span
Bracket s Minimum of three at top, 5-point pattern over area of
middle and bottom of tank bracket
Horizontal stringers All stringers with 5-point pattern over one
measurements at both ends square metre area, plus single
and middle measurement s near bracket
toes and on flanges
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS

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SOURCE: MSC.197(80).pdf
and middle measurement s near bracket
toes and on flanges
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 53 -
ANNEX 5
MINIMUM REQUIREMENTS FOR OVERALL AND CLOSE-UP SURVEY AND THICKNESS MEASUREMENTS AT INTERMEDIATE SURVEY OF DOUBLE HULL OIL TANKERS
5<age< 10 years 1 O<age< 15 years Age > 15 years
1 2 3
Overall survey of Overall survey of all As for renewal survey given
representative salt water salt water ballast tanks, in annex 1
ballast tanks, selected by the including combined
attending surveyor (the cargolballast tanks